



國立陽明交通大學
NATIONAL YANG MING CHIAO TUNG UNIVERSITY

COURSE: Robotics

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Project 2: Motion planning

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1 Forward Kinematics

As a reminder the matrix is:

$${}^0_6T = \begin{pmatrix} r_{1,1} & r_{1,2} & r_{1,3} & p(x) \\ r_{2,1} & r_{2,2} & r_{2,3} & p(y) \\ r_{3,1} & r_{3,2} & r_{3,3} & p(z) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The individual elements derived from the kinematic chain are:

$$\begin{aligned} r_{1,1} &= c_1(c_{2,3,4}c_5c_6 - s_{2,3,4}s_6) - s_1(-s_5c_6) \\ &= c_1c_{2,3,4}c_5c_6 - c_1s_{2,3,4}s_6 + s_1s_5c_6 \end{aligned}$$

$$\begin{aligned} r_{1,2} &= c_1(-c_{2,3,4}c_5s_6 - s_{2,3,4}c_6) - s_1(s_5s_6) \\ &= -c_1c_{2,3,4}c_5s_6 - c_1s_{2,3,4}c_6 - s_1s_5s_6 \end{aligned}$$

$$r_{1,3} = -c_1s_5c_{2,3,4} + s_1c_5$$

$$p(x) = c_1(a_2c_2 + a_3c_{2,3} + d_5s_{2,3,4} - d_6s_5c_{2,3,4}) + s_1d_4 + s_1d_6c_5$$

$$r_{2,1} = s_1c_{2,3,4}c_5c_6 - s_1s_{2,3,4}s_6 - c_1s_5c_6$$

$$r_{2,2} = -s_1c_{2,3,4}c_5s_6 - s_1s_{2,3,4}c_6 + c_1s_5s_6$$

$$r_{2,3} = -s_1s_5c_{2,3,4} - c_1c_5$$

$$p(y) = s_1(a_2c_2 + a_3c_{2,3} + d_5s_{2,3,4} - d_6s_5c_{2,3,4}) - c_1d_4 - c_1d_6c_5$$

$$r_{3,1} = s_{2,3,4}c_5c_6 + c_{2,3,4}s_6$$

$$r_{3,2} = -s_{2,3,4}c_5s_6 + c_{2,3,4}c_6$$

$$r_{3,3} = -s_5s_{2,3,4}$$

$$p(z) = a_2s_2 + a_3s_{2,3} - d_5c_{2,3,4} - d_6s_5s_{2,3,4} + d_1$$

2 Inverse Kinematics

The inverse kinematic gave us:

$$\begin{cases} \Delta < 0 & \text{no solution } (c_5 \notin [-1, 1]) \\ \Delta = 0 & \theta_5 = \pm \cos^{-1} \left(\frac{-B}{2A} \right) \\ \Delta > 0 & \theta_5 = \pm \cos^{-1} \left(\frac{-B \pm \sqrt{\Delta}}{2A} \right) \end{cases}$$

$$\theta_1 = \text{atan2} \left(\pm \sqrt{1 - \left(\frac{-d_4 - c_5 d_6}{r} \right)^2}, \frac{-d_4 - c_5 d_6}{r} \right) + \text{atan2}(-p_x, p_y)$$

$$\theta_6 = \text{atan2} \left(\frac{1}{s_5}(c_1 r_{2,2} - s_1 r_{1,2}), \frac{1}{s_5}(s_1 r_{1,1} - c_1 r_{2,1}) \right)$$

After estimating θ_4 we have:

$$\theta_2 = \text{atan2}(k_2 k_3 - k_1 k_4, k_1 k_3 + k_2 k_4)$$

$$\theta_3 = \text{atan2} \left(\pm \sqrt{1 - \left[\frac{1}{2a_2 r_3} (K_1^2 + K_2^2 - K_5^2 - K_6^2 - a_2^2) \right]^2}, \frac{1}{2a_2 r_3} (K_1^2 + K_2^2 - K_5^2 - K_6^2 - a_2^2) \right) + \text{atan2}(-K_6, K_5)$$

3 Motion planning

3.1 Implementation choices

For this motion planning problem multiple choices are made:

- The first and most important one is the usage of **joint space**.
This space prevents the appearance of singularities and so highly simplifies the problem.
It also drastically reduces the calculation time as inverse kinematic which in our case is time consuming.
While reducing the computation of inverse kinematics it also imply reducing the amount of choices needed in case of multiple solutions.

- The second choice is the method of selection in case of multiple solutions of the inverse-kinematic.

To make this choice let's define:

$$C(\Theta, \Theta') = \sum_{k=1}^6 (7 - k)(\Theta_k - \Theta'_k)^2$$

Here the idea is to use C as a cost function to estimate the cost of a movement from Θ to Θ' .

Θ and Θ' being vectors of joint angles. The cost here is a sort of weighted distance making a movement of the base more important than one of the wrist.

We then obviously choose the movement with the lowest cost.

In our problem we will choose the cheapest combination of movement from A to B and then C .

Hence we get:

$$\dot{\Theta}_{BC} = \frac{\Theta_C - \Theta_B}{t_{dBC} - \frac{1}{2}t_C} = \frac{\Theta_C - \Theta_B}{t_{dBC} - \frac{1}{2}t_{acc}}$$

$$\dot{\Theta}_{AB} = \frac{\Theta_B - \Theta_A}{t_{dAB} - \frac{1}{2}t_{acc}}$$

Now we need the acceleration during the different transition. We will consider the accelerations constant. $\dot{\Theta}_A = 0$ and $\dot{\Theta}$ is continuous. So $\dot{\Theta}_A = \dot{\Theta}(0) = 0$, $\dot{\Theta}_{AB} = \dot{\Theta}(t_A)$.

And:

$$\ddot{\Theta}_A = \frac{\dot{\Theta}_{AB}}{t_A}$$

Using the same logic:

$$\ddot{\Theta}_C = \frac{-\dot{\Theta}_{BC}}{t_C}$$

And:

$$\ddot{\Theta}_B = \frac{\dot{\Theta}_{BC} - \dot{\Theta}_{AB}}{2t_{acc}}$$

Finally we get:

$$\Theta(t) = \begin{cases} \Theta_A + \frac{t^2}{2}\ddot{\Theta}_A & \text{if } t \leq t_A \\ \Theta(t_A) + (t - t_A)\dot{\Theta}_{AB} & \text{if } t_A < t \leq t_A + t_{AB} \\ \Theta(t_A + t_{AB}) + (t - t_A - t_{AB})\dot{\Theta}_{AB} + \frac{(t - t_A - t_{AB})^2}{2}\ddot{\Theta}_B & \text{if } t_A + t_{AB} < t \leq t_A + t_{AB} + t_B \\ \Theta(t_A + t_{AB} + t_B) + (t - t_A - t_{AB} - t_B)\dot{\Theta}_{BC} & \text{if } t_A + t_{AB} + t_B < t \leq t_A + t_{AB} + t_B + t_{BC} \\ \Theta(t_A + t_{AB} + t_B + t_{BC}) + \frac{(t - t_A - t_{AB} - t_B - t_{BC})^2}{2}\ddot{\Theta}_C & \text{if } t > t_A + t_{AB} + t_B + t_{BC} \end{cases}$$

3.3 Pipeline

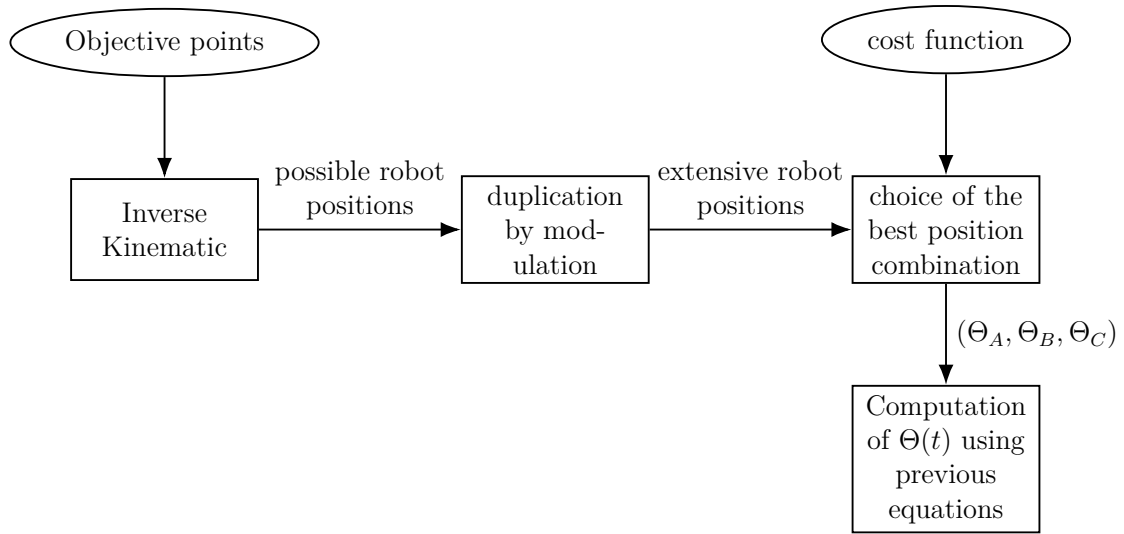


Figure 1: Motion Planning Pipeline

4 Results

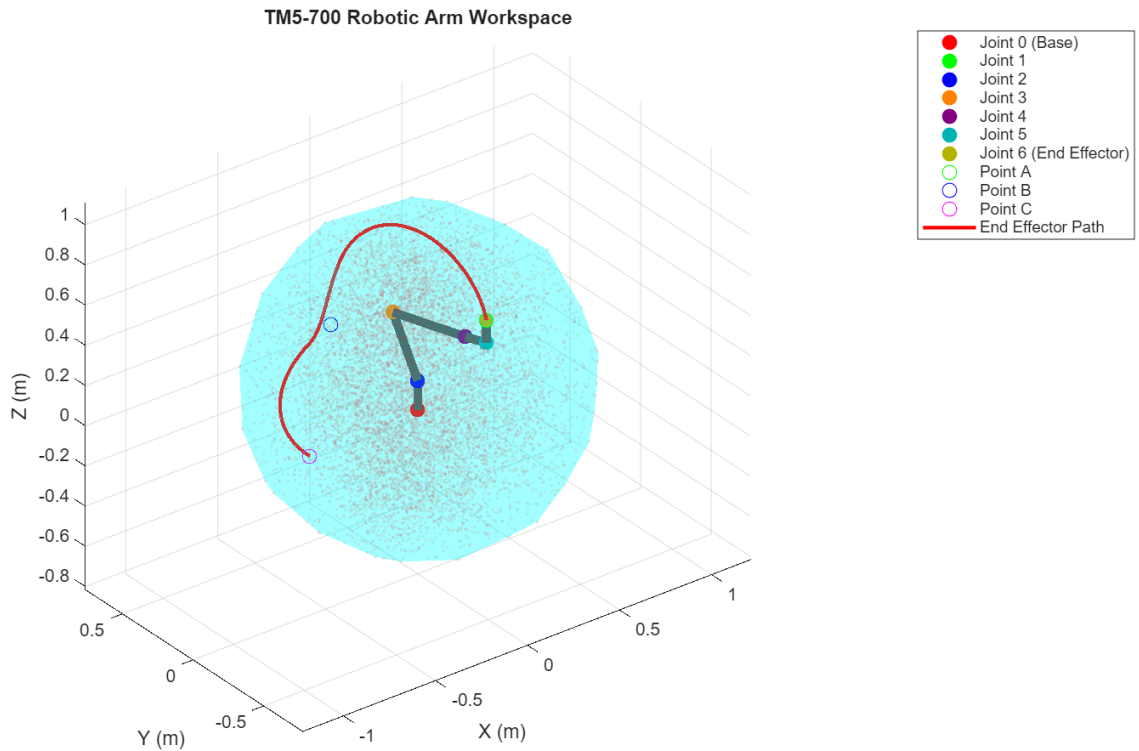


Figure 2: End effector path

The video of the motion is linked in this zip file.

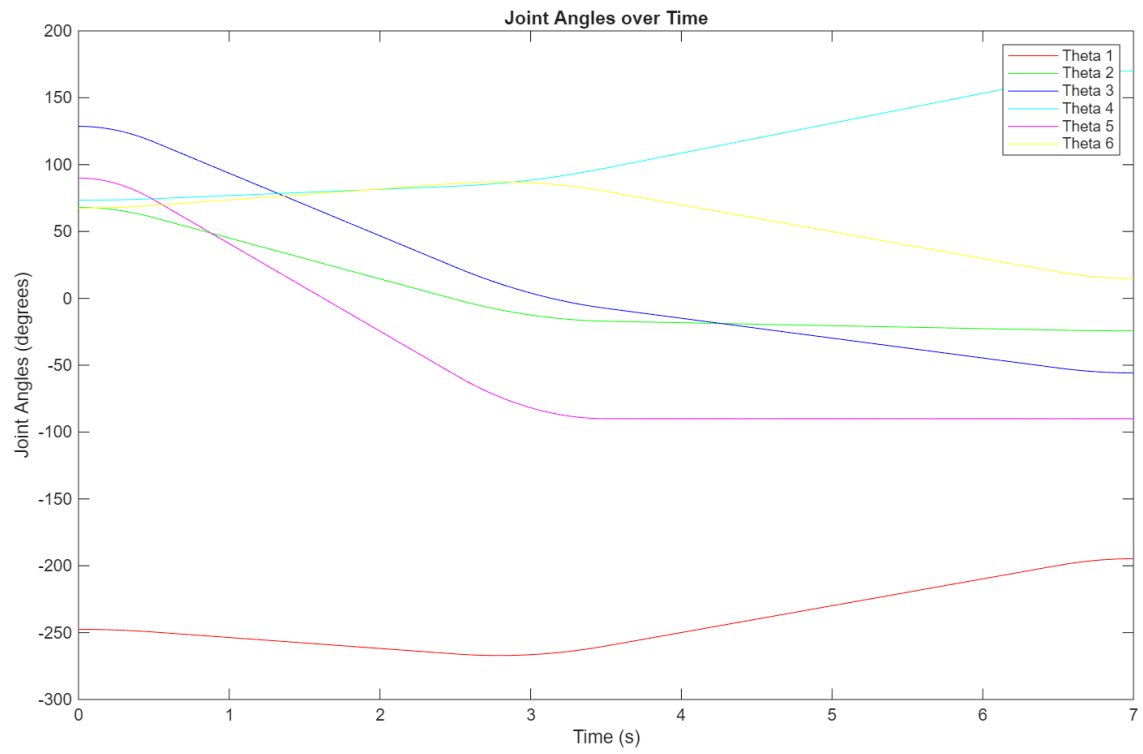


Figure 3: Joint angles over time

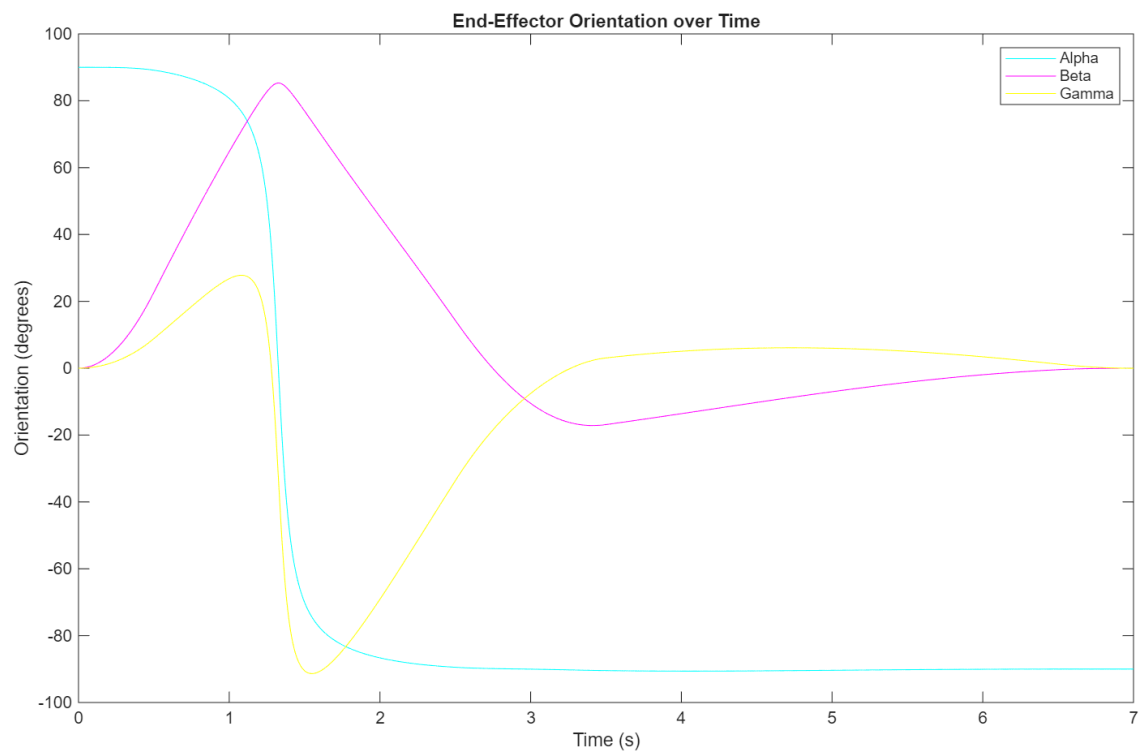


Figure 4: End effector orientation over time

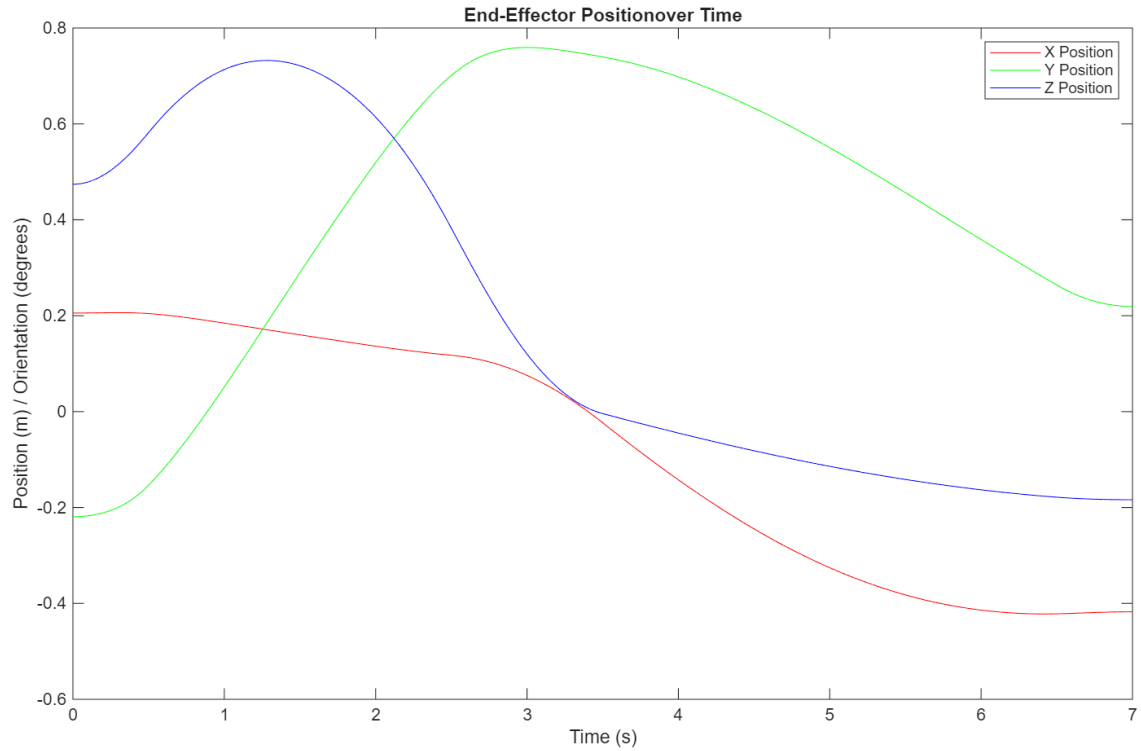


Figure 5: End effector position over time

We can easily observe here that 2 straight lines are being formed in joint space but not in cartesian. It is totally logical considering the fact that we use joint space for the planning.

We also see continuity without spiking in joint angles which suggest the continuity of the joint angles speeds.

References

- [1] J.-C. Hong, *Robotics Course Lecture Notes*, National Yang Ming Chiao Tung University, Fall 2025.